Sora Kim

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Research Focus

I study how regulated work environments shape practitioner cognition, decision-making, and accountability—a space I call **Regulatory HCI**. The core question: how do these environments quietly determine what workers can see, do, and be held accountable for—and what happens when AI enters that design? Grounded in four years of embedded fieldwork in pharmaceutical Good Manufacturing Practice (GMP)—the FDA-regulated environment governing pharmaceutical manufacturing—I use whatever the question requires: ethnographic observation, organizational theory, STS, NLP and corpus analysis, or prototype interfaces. The methods follow the question.

Keywords: Regulatory HCI · AI in regulated work · practitioner cognition · judgment preservation · sociotechnical systems · computational text analysis

Education

Chungbuk National University *Cheongju, South Korea*
B.A. in English Literature and Linguistics March 2015 – August 2022

- **Top-Ranked Admission**, Humanities & Social Sciences Division, 2015 (entrance examination).
- **Full Merit Scholarship** (tuition & fees, 8 semesters; merit-based).

Self-Directed Study *Remote*
Programming, Data Science, Machine Learning March 2021 – August 2022

- 7-course specialization in data science and computational methods (Johns Hopkins, Coursera, 2021): algorithms, statistics, Python, R, command-line tools applied to large-scale datasets—foundation for later corpus analysis.
- Applied deep learning projects: image classification (PyTorch, CNN); unsupervised learning; ensemble methods.

Industry Research Experience

GMP-Regulated Pharmaceutical Manufacturing Site *Seoul, South Korea*
Quality Control / Quality Systems September 2022 – Present

- Embedded practitioner-researcher in regulated biomanufacturing quality operations (~4 years); sustained ethnographic observation of Standard Operating Procedure (SOP) use, deviation handling, Corrective and Preventive Action (CAPA) processes, and FDA audit workflows.

- Designed and deployed 9 productivity tools serving Quality Control (QC) operations (see Deployed Systems); observational data and fieldwork form the empirical foundation of the research program below.
- Designed and delivered a differentiated AI training program for the QC team (managers: tool-building; operators: AI-assisted authoring), motivating the educational artifacts below.

Deployed Systems *(Empirical Substrate for Design Research)*

Production tools deployed at a GMP-regulated pharmaceutical site, serving as the empirical substrate for the design research below. All systems operate as unvalidated productivity layers external to validated GMP systems, with team lead awareness and support.

- **QC Stability Testing Dashboard** — Deadline-tracking interface for stability testing programs; team-wide adoption, ongoing operation.
- **SOP Retrieval System** — Semantic search tool for GMP procedural documents; empirical base for *From Retrieval to Reasoning*.
- **Document Navigation Tool (Traceability Layer)** — Cross-document linking interface with metadata mapping across regulated quality documentation; empirical base for *When Documents Don't Connect* (traceability component).
- **Navigation Behavior Tracker** — Behavioral tracking layer capturing navigation paths, hesitation, and cross-reference jumps in regulated document work; empirical base for *When Documents Don't Connect* (loopback component).
- **Review Dashboard** — Prioritized document review interface surfacing procedural friction before formal deviation; empirical base for *Friction Before Failure*.
- **Root Cause Analysis (RCA) Scaffolder** — Investigative scaffold that preserves analyst judgment while structuring causal reasoning; empirical base for *Keeping Comparison Open*.
- **Veeva Vault Change Control Assistant** — Workflow support for electronic change control submissions.
- **Reagent Inventory & Expiry Management Tool** — Automated expiry tracking and inventory alerts for QC reagents.
- **Document Auto-Review AI** — Format and compliance pre-check for procedural documents.

Publications

Organizational Theory & STS

Four-paper lifecycle theory: pre-action structuring (AAC), procedural document infrastructure (CF), investigative convergence (RC), AI-accelerated closure (PAC).

Kim, S. (2026). *Anticipated Accountability Convergence: How Organizations Structure Responsibility Before Action.*

Kim, S. (2026). *Cognitive Friction: Procedural Documents as Accountability Infrastructure in Regulated Organizations.* [Companion code: sorakim-lab/sop-cognitive-friction-pilot.]

Kim, S. (2026). *Responsibility Convergence: How Investigative Systems Stabilize Attribution.*

Kim, S. (2026). *Premature Accountability Convergence: How Technology Accelerates Responsibility*

Stabilization in Organizations. [Companion code: sorakim-lab/fda-warning-letters-pac-analysis.]

HCI Design Research

Five design studies in a four-layer framework (accessibility, structure, coordination, reasoning) for Regulatory HCI; each grounded in a deployed tool (see Deployed Systems), with N=20 user studies on five prototypes.

Kim, S. (2026). *Accountability Infrastructure Design: A Four-Layer Framework for Judgment-Preserving Systems in Regulated Work.*

Kim, S. (2026). *Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations.*

Kim, S. (2026). *Friction Before Failure: Making Review Burden Visible in Regulated Document Work.*

Kim, S. (2026). *From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work.*

Kim, S. (2026). *When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation.*

Computational Social Science

Computational program operationalizing accountability across multiple regulatory domains and method layers. Cross-domain replication validates the lifecycle theory developed on FDA data.

Kim, S. (2026). *Designed Attribution: Coding-Level and Grammatical-Level Evidence of Organizational Accountability Displacement in NTSB Aviation Investigations.* [Replication code: sorakim-lab/ntsb-attribution-analysis. Two-phase analysis: structured coding (n=71,114 findings) and dependency-parsed probable cause statements (n=23,742). Triangulated finding: organizational accountability appears in 0.42–0.58% of NTSB outputs across both methods.]

Kim, S. (2026). *Responsibility as Multi-Layer Signal: A Computational Framework for FDA Warning Letters.* [Synthesizes four computational components, each with a companion repository: premature-convergence-detection, accountability-graph-extraction, responsibility-attribution-prediction, responsibility-distribution-metric.]

Educational Artifacts *(Open Access)*

Public at sorakim-lab.github.io/teaching.

- ***How to Actually Use Claude: A GMP Quality Lab AI Literacy Series*** (2026) — A 15-page AI literacy cookbook for pharmaceutical GMP teams. Four chapters covering AI capabilities and limitations; security and data integrity (no credentials, no internal regulated systems such as Veeva Vault, LIMS, or shared network drives; ALCOA+ data integrity standards); document writing prompts (SOP, deviation reports, CAPA, translation); and workflow automation for managers, plus a quick-reference prompt card. *Bilingual (English + Korean)*; PDF.
- **DAY ONE: GMP** (2026) — A pixel-art onboarding game for newcomers entering pharmaceutical GMP environments. Five workday scenarios (gowning, material verification, SOP retrieval

under pressure, deviation reporting, end-of-day sign-off) teach core principles of regulated practice through structured decision points and explanation-after-choice feedback. Built in HTML + JavaScript (Canvas API) with retro pixel typography. *Bilingual (English + Korean)*; web.

Conferences

- Kim, S. (2026). *Premature Accountability Convergence: Inscription, Closure Pressure, and the Stabilization of Responsibility.* Abstract submitted to **4S (Society for Social Studies of Science) Annual Meeting**, Toronto, 2026 (Panel 74).

Research Corpus & Data Infrastructure

- **FDA Warning Letters Corpus** — N=50 letters, 160 violations (2016–2025). FDA Warning Letters are the agency's formal regulatory enforcement documents issued when significant compliance failures are identified; used across organizational theory, STS, and computational papers.
- **NTSB (National Transportation Safety Board) Aviation Accident Corpus** — Full Aviation Accident Database (*avall.mdb*, ~546 MB; 71,114 findings, 27,877 narratives, 1982–2025) directly analyzed; replication code at sorakim-lab/ntsb-attribution-analysis. Methodological contrast with FDA Warning Letters: pharmaceutical regulation makes only final outputs public, leaving the investigative process unobservable; aviation regulation makes the entire investigative pipeline traceable. The cross-domain comparison provides the strongest external validation site for the lifecycle theory developed on FDA data.
- **Cognitive Friction Corpus** — 400 naturally occurring sentences from regulated procedural documents, manually annotated at sentence level.
- **Field Observation Archive** — Nearly four years of notes on SOP use, deviation handling, and CAPA in a GMP-regulated site.

Methods & Technical Skills

- **Research Methods** — Ethnographic fieldwork; document & corpus analysis; within-subject experimental design; think-aloud protocols; behavioral logging; qualitative coding; annotation protocol design; causal reasoning analysis; statistical analysis.
- **Programming & Development** — Python, R; Streamlit (deployment); JavaScript, HTML/CSS; Git/GitHub; LaTeX.
- **Text Analysis & NLP** — TF-IDF; sentence-level annotation; document classification; corpus construction; entity linking; temporal embedding; graph extraction; spaCy dependency parsing; rule-based actor extraction.
- **Machine Learning** — PyTorch (CNN, deep learning for image classification); scikit-learn; classification; ensemble methods; unsupervised learning.
- **Regulated-Environment Systems** — Veeva Vault Quality Management System (industry-standard pharma QMS); GMP documentation systems; electronic change control; deviation/CAPA workflows.

Languages

Korean (native) · English (fluent) · Japanese (fluent)